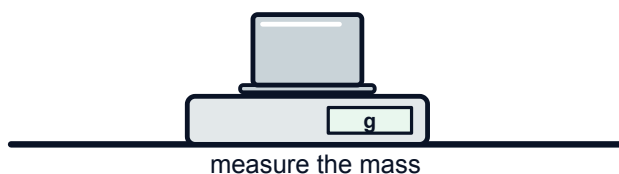


Density of a regular solid

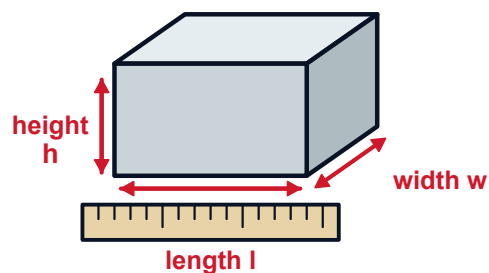
A student measures the density of a rectangular metal block.

- 1 Measure the mass of the block with a balance.

metal block
mass $m = \text{___ g}$



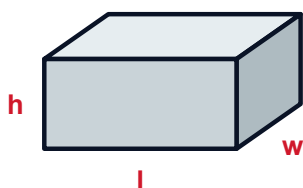
- 2 Measure the length, width and height of the block with a ruler.



- 3 Calculate the volume: length $\times$ width $\times$ height.

$$V = l \times w \times h$$

volume (cm^3) = length $\times$ width $\times$ height



- 4 Calculate the density: mass $\div$ volume.

$$\rho = m \div V$$

density (g/cm^3) = mass (g) $\div$ volume (cm^3)

use m from step 1 and V from step 3

Note: The two measuring steps can be done in either order.